

Apply for a Science Corps Fellowship - Entry #3230

Name

Kathy Wen

Emailkathy.wen@spc.ox.ac.uk**Gender**

Female

City, state, and country of residence

Oxford

Nationality

Australia

When did or will you finish your PhD?

December 2023

University Education

DPhil Engineering Science University of Oxford 2023

BE (Hons) Civil Engineering University of New South Wales 2018

What are your reasons for applying to be a Science Corps fellow?

The motivation behind applying for this fellowship follows from my commitment in acting to reduce global inequality and encouraging sustainable development. Sustainable development is the focus of my ongoing research in green technology, which is targeted at optimising design methods for offshore wind turbines. The aim of this research is to ensure the economic viability of offshore wind as a renewable energy source, enabling global targets of CO2 emissions to be achieved. I frequently showcase my work by delivering academic workshops that explain sustainable engineering principles to high school students as a part of St Peter's College Outreach program. Outside my technical experience and training, I actively volunteer in

the not-for-profit sector for organisations such as The St Vincent De Pauls, and Jacari. My main contribution within these organisations has been in education, through providing free tutoring services for students of refugee and immigrant backgrounds. This experience has led me to understand that education can inspire and empower and has reiterated the importance of access to education as a fundamental human right. I am looking for opportunities to utilise my expertise to solve global challenges and it is an environment such as Science Corps where I believe this can occur.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

I have extensive experience teaching engineering courses at various educational levels and institutions. At the tertiary level, I've instructed courses like geotechnical engineering and fluid mechanics for undergraduate students at the University of Oxford and the University of New South Wales. These courses included Soil Mechanics, Geotechnics, and CVEN3203: Applied Geotechnics.

Additionally, I've served as an academic tutor within Regent's College – Oxford Prospects and Global Development Institute (OPDGI). In this role, I've created and conducted engineering seminars focused on academic skills such as reading, writing, and presenting in engineering science. I've also designed and delivered outreach seminars like "Introduction to Sustainable Development" and "Beyond Simple Numbers: Introduction to Bayesian Statistics" to underprivileged high school students across the UK at St. Peter's College, Oxford. These seminars were crafted to foster a deeper understanding of scientific complexities and empower students to develop innovative solutions for today's challenges. The fast-paced nature of these teaching positions has sharpened my ability to effectively communicate scientific theory through relevant analogies, clear examples, and practical problem-solving which I feel are relevant for the Science Corps fellowship.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

In science education

In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

In the near future, I aim to pursue a post-doctoral research position in academia, focusing on advancing renewable energy and offshore wind turbine foundation design. My current research background has honed my ability to identify complicated problems from literature and employ experimental and programmatic methods in which to solve them as evidenced in my research papers. Simultaneously, I'm committed to public engagement. I plan to continue participating in academic outreach programs, particularly those targeting underserved communities, to bridge the gap between science and society. Looking ahead to the long term, I envision myself as a lecturer or associate professor at an academic institute. In this role, I will have the opportunity conduct impactful research while also being a part of an academic community that fosters learning and knowledge sharing. Additionally, I will have the privilege of educating and inspiring students. By nurturing their passion for science and sustainability, I hope to contribute significantly to the field while fostering a new generation of enthusiastic learners.

CV/Resume[20230904-CV.pdf](#)**How did you find out about Science Corps?**

University email list

Any other comments/questions?

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